Camera-Proxy Enhanced Identity-Recalibration Learning for Unsupervised Visible-Infrared Person Re-Identification

Run-Sen Xia, Xue-Yan Wang, Si-Bao Chen, *Member, IEEE*, Jin Tang, and Bin Luo, *Senior Member, IEEE*

***Abstract*— Visible-Infrared person Re-Identification (VI-ReID) involves querying images of the same person across visible and infrared modalities. To minimize annotation costs, Unsupervised Visible-Infrared person Re-Identification (UVI-ReID) using pseudo-label contrastive learning has emerged. Traditional UVI-ReID approaches often neglected camera domain information and relied on inadequate update strategies during training, only using cosine distance for testing, which led to incorrect mapping of cross-modal relationships. To address these issues, we propose Camera-proxy Enhanced Identity-recalibration Learning (CEIL). It consists of two main stages: first, it employs intra-modal contrastive learning in conjunction with the camera-proxy, updates the memory bank using our innovative Difficulty-aware Cluster-based Memory Updating (DCMU) strategy, and applies Camera Domain-driven Local correlation (CDL) Loss to enhance the learning process. Then utilizes cross-modal contrastive learning, featuring our Proxy-enhanced Cross-modal Mapping (PCM) module, to recalibrate the identity relationships between different modalities. Graph network-based Camera constraint adjustment Re-ranking (GCR) method is adopted during test, utilizing camera domain information to recalibrate the correspondence between identities. Extensive experiments have demonstrated that CEIL achieving state-of-the-art performance on the SYSU-MM01, RegDB, and LLCM datasets and the GCR, as a general unsupervised re-ranking method, can further enhance performance of model on these datasets. The code will be released at https://github.com/maybeextra/CEIL.**

***Index Terms*— Cross-modal contrastive learning, person re-identification, camera-proxy, identity-recalibration.**

I. Introduction

In RECENT years, the demand for intelligent video surveillance has driven progress in person Re-Identification (ReID) [1], [2], [3], [4], which matches pedestrians across non-overlapping camera views. While supervised and unsupervised methods [5], [6], [7], [8] excel in well-lit environments, they falter in low-light or nighttime conditions due to limited information. Efforts such as optimizing loss functions [9], [10], [11] have been made, but single-modal approaches remain inadequate, necessitating more universal solutions.

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The authors are with the MOE Key Laboratory of ICSP, the IMIS Laboratory of Anhui, Anhui Provincial Key Laboratory of Multimodal Cognitive Computation, and the Zenmorn-AHU AI Joint Laboratory, School of Computer Science and Technology, Anhui University, Hefei 230601, China (e-mail: 2966461966@qq.com; 518049@qq.com; sbchen@ahu.edu.cn; tj@ahu.edu.cn; luobin@ahu.edu.cn).

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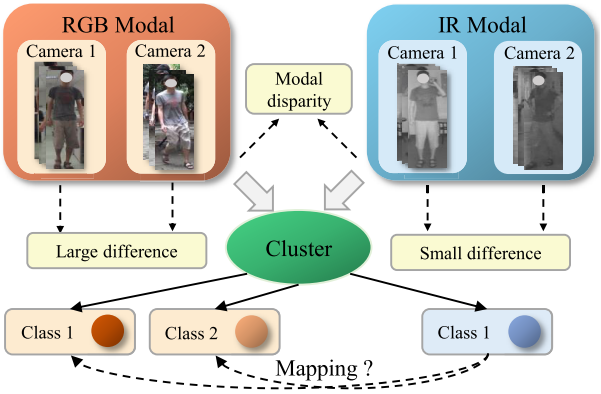

Fig. 1. Illustration of the dilemma. Images of the same identification exhibit significant differences across multiple modalities, and within each modality, variations arise due to different camera positions and times. This ultimately leads to inconsistent clustering and ambiguous cross-modal label association.

With the development of technology, recently deployed cameras can seamlessly transition to far or near-infrared mode during nighttime operations [12]. Taking advantage of this capability, alongside the development of extensive cross-modal datasets [13], [14], [15] annotated by humans, various techniques [5], [16], [17], [18], [19], [20], [21], [22], [23] have been proposed for Visible-Infrared person Re-identification (VI-ReID). In contrast to the annotation of single-modal ReID datasets, annotating cross-modal datasets demands considerably more resources. It is undeniable that the current high-performing VI-ReID methods heavily rely on annotated training datasets. In response to this challenge, Unsupervised Visible-Infrared person Re-Identification (UVI-ReID) methods [24] offers a promising solution.

Existing UVI-ReID methods focus on unsupervised contrastive learning using pseudo-labels generated by clustering. However, this learning strategy still has flaws. The visible-infrared image space is characterized by substantial modal gaps and pedestrian images within the same modality exist high intra-Identification variance and inter-Identification similarity. This resulted in suboptimal modal cohesive classes and incorrect cross modal mappings as shown in Fig.1.

To tackle these challenges, it's important to reassess the UVI-ReID process. Variations in image sources, even within the same modality, can complicate identification. Fortunately,

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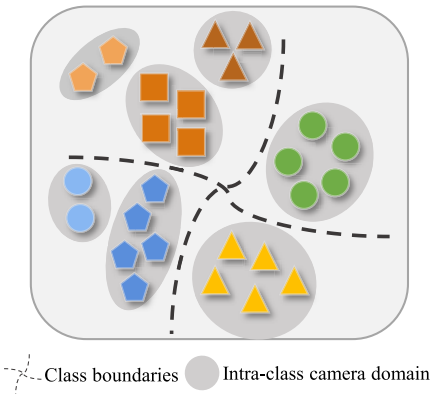

Fig. 2. Illustration of using camera information to segment the generation of pseudo-labels. The four sections delineated by class boundaries represent the distinct categories generated by DBSCAN clustering. On this foundation, camera information is used to further divide and differentiate the camera proxies. In the diagram, different shapes represent different cameras, and each intra-class camera domain encapsulates a distinct camera proxy.

acquiring camera domain information is easier than obtaining true labels. It allows clusters to be subdivided into camera proxies, as shown in Fig.2. Re-ID methods like ICE [25] use this approach, and DCCL [26] applies it to UVI-ReID with camera-aware and class-based contrastive learning. However, introducing too many training objectives when the foundation is unstable can be burdensome. Cross-modal identity learning faces similar challenges, as successful cross-modal clustering relies on reliable intra-modal clustering first. Therefore, we propose a two-stage Camera-proxy Enhanced Identity-recalibration Learning (CEIL) framework, depicted in Fig.3.

To effectively integrate camera proxies throughout the training process while minimizing interference from excessive information, we employ a two-stage approach. In the first stage, we conduct contrastive learning exclusively at the camera proxy-level. We introduce a Camera Domain-driven Local correlation (CDL) Loss. In single-modal scenarios, camera variations mainly stem from lighting and position changes. However, in UVI-ReID, visible and infrared cameras exhibit distinct sensor artifacts like noise patterns and thermal drift. It's crucial to reduce camera domain interference and compute losses separately for each domain. Leveraging camera information not only addresses single-modal issues but also provides a solid foundation for cross-modal learning. Additionally, local loss computation slows training, mitigating negative impacts from clustering errors. In the second stage, we incorporate a class-level loss based on pseudo-labels. Drawing inspiration from PGM [27], we develop a Proxy-enhanced Cross-modal Mapping (PCM) module. This module employs multi-level contrastive learning to bridge modality gaps, according to dual consistency [26] in cross-modal correspondences between camera proxies and classes, to help correct errors and enhance performance. In single-modal scenarios, camera information mainly aids cross-camera feature alignment. For UVI-ReID tasks, the primary challenge becomes cross-modal matching, as category-level matching alone is often unstable. By incorporating camera information as auxiliary features, a camera proxy hierarchy can be established beyond the category level, enabling the construction of cross-modal matching graphs both within proxies and categories. This approach significantly differs from traditional single-modal uses of camera information. The multi-level mapping consistency mechanism [26] across categories and camera proxies effectively mitigates modal discrepancies, improving the accuracy and robustness of cross-modal matching. Traditional methods such as those described in AGW [5], which use momentum updates with memory banks for contrastive learning, assume spherical sample distributions. This assumption conflicts with reality. In fact, clusters generated by clustering algorithms often exhibit a ring-shaped distribution. While RTMem [28] introduced real-time memory updates, it overlooked challenging samples. To address this, our Difficulty-aware Cluster-based Memory Updating (DCMU) strategy dynamically adjusts based on the relationship between samples and proxy memory, ensuring more effective learning. Upon completing the two-stage training, re-ranking can significantly improve ReID performance, as shown in studies like [29] and [30]. However, its reliance on CPU computation can be time-consuming, reducing its application in recent research. Although GCN [31] enhances speed, it lacks the integration of prior camera information, often prioritizing speed over accuracy. To overcome these limitations, we propose a Graph network-based Camera constraint adjustment Re-ranking (GCR) method. This method combines camera information with features and utilizes GPU-accelerated graph convolution, significantly reducing computation time while maintaining strong performance.

Our main contributions can be summarized as follows:

- Proposed the CDL Loss, using local negative samples within the camera domain to improve domain consistency, generalization, and discriminative ability.
- Introduced the PCM module to align multimodal clusters at class and camera levels, improving robustness.
- Developed the DCMU strategy, updating the memory bank based on data difficulty to enhance performance.
- Created the GCR method to integrate camera information and perform graph propagation, thereby significantly reducing time waste and achieving a leap in performance.

Extensive experiments on the SYSU-MM01, RegDB, and LLCM datasets demonstrate that the CEIL achieves state-of-the-art (SOTA) performance compared to recent methods.

II. Related Work

A. Contrastive Learning in Person Re-Identification

Contrastive Learning serves as a technical basis for pseudo-label-based unsupervised person ReID, attracting considerable attention for its critical role. This approach learns discriminative feature representations through contrastive optimization in the feature space. It constructs positive and negative sample pairs, simultaneously minimizing the distance between positive pairs while maximizing the separation of negative pairs.

Advancements in contrastive learning have significantly propelled the development of unsupervised person ReID. The seminal work of SimCLR [32] established a breakthrough

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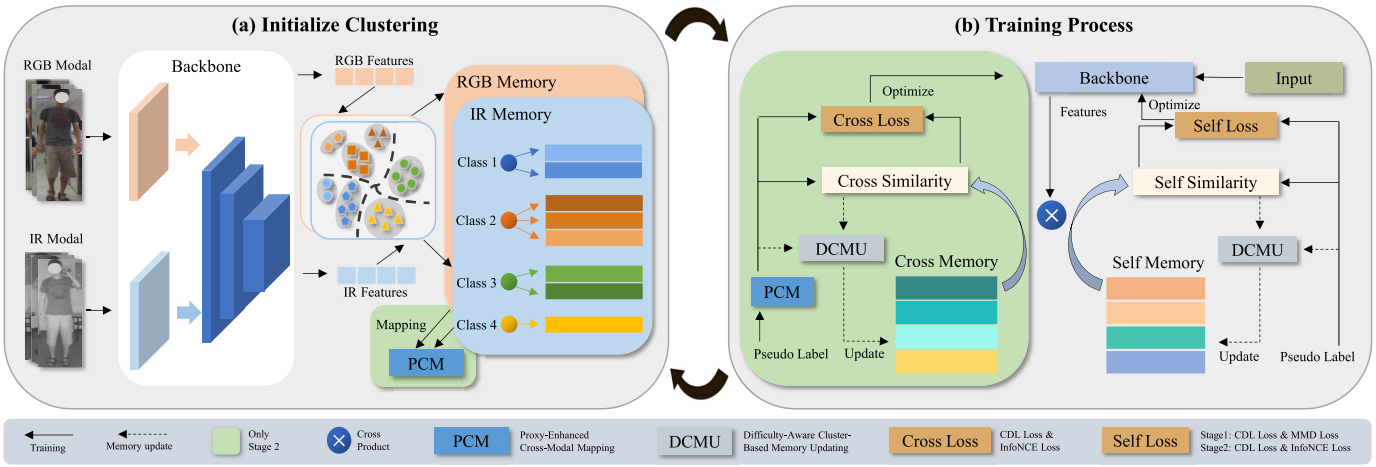

Fig. 3. The overall architecture of the proposed CEIL. Includes a difficulty-aware cluster-based memory update (DCMU) strategy, an proxy-enhanced cross-modal mapping (PCM) module, and a camera domain-driven local correlation (CDL) loss. At the start of each epoch, clustering is initialized, and clusters are divided by camera source for contrastive learning. During the first training stage, CDL loss is applied within the same modality based on camera agents, and MMD loss is applied between modalities. In the second stage, not only CDL loss is applied within modalities, but also InfoNCE loss is applied based on clusters, and the PCM is used for cross-modal feature learning, with an equivalent loss calculated in the cross modality for each sample.

framework by integrating data augmentation, non-linear projection heads, and large-batch training to learn visual representations without supervision, with its core principle being the maximization of similarity between augmented views of the same images. Parallelly, MoCo [33] addressed the limitations of negative samples through its innovative momentum encoder and dynamic dictionary, thereby enhancing feature consistency. These foundational contributions paved the way for modern pseudo-label-based unsupervised person ReID methodologies. Building upon these principles, SSL [34] pioneered the application of contrastive learning in unsupervised person re-identification by optimizing feature similarity through positive and negative sample pairs. This approach was further refined by SpCL [35], which incorporated self-paced learning and a hybrid memory mechanism, marking substantial progress in the field. SSFT [14] extended these concepts by proposing a novel framework for UVI-ReID, achieving remarkable improvements in cross-modality matching performance through shared-specific feature learning and self-paced contrastive learning, and established the theoretical and methodological foundation for our proposed CEIL framework.

B. Visible-Infrared Person Re-Identification

1) Supervised Methods: The issue of visible-infrared cross-modal pedestrian re-identification was initially introduced by Wu et al. [36]. Ye et al. [5] later introduced the comprehensive SYSU-MM01 dataset, establishing a standardized benchmark with the AGW method and proposing a novel evaluation metric called mINP. Supervised VI-ReID [17], [37], [38], [39] has focused on bridging the gap between the visible and infrared modalities and learning robust representations to effectively handle these modal differences. GECNet [40] proposes a grayscale enhancement coloring network that compensates for the modality gap by preserving the information-rich structures contained in color images. Methods such as MAUM [17] and MSCLNet [18] have concentrated on learning modality-specific features and integrating them to reduce noise and enhance identity discrimination. PMT [20] and CycleTrans [21], introduced auxiliary modalities and lightweight modules to capture richer semantic information. Further advancements such as PartMix [19] and DMA [22] have leveraged data augmentation techniques and optical characteristics to mitigate cross-modal differences and preserve identity features. HOS-Net [23] has focused on learning a common feature space by using aligned features from different modalities. These advancements not only improved the performance of supervised VI-ReID but also provided valuable insights for developing unsupervised methods in this domain.

2) Unsupervised Methods: UVI-ReID remained an area of extensive research, focusing on bridging the gap between visible and infrared modalities through innovative approaches. OTLA [41] utilized optimal transport for pseudo label assignment across modalities. ADCA [24] enhanced inter-modality representation learning via a dual-contrastive framework. Recent works in unsupervised visible-infrared person re-identification have focused on bridging the modality gap and learning modality-invariant features through various cross-modal alignment and clustering strategies. Methods like CCLNet [42] and Sun et al. [43] leveraged CLIP to generate cluster-aware prompts and extract deep text information, facilitating the creation of text-image alignment modules that bridged the visible-infrared gap. Approaches such as PGM [27], MMM [44], and RPNR [45] established reliable cross-modality correspondences through progressive graph matching, cross-modality clustering, and optimal transport prototype matching, respectively. Several works, including CHCR [46] and MIMR [47], focused on refining or eliminating noisy pseudo-labels by analyzing clustering results or refining labels in ambiguous samples. Mutual learning and modality-specific information extraction were addressed by methods like CMAM [48] and PCAL [49], which incorporated cross-modal asymmetric mutual learning algorithms and modality-invariant contrastive learning to reduce modal differ-

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ences and learn invariant representations. Additionally, Teng et al. [50] and Shi et al. [51] proposed strategies to preserve cross-modal correlations and used progressive contrastive learning with dynamic prototypes to establish relationships between different modalities. GUR [52] first introduced camera proxies into UVI-ReID, systematically exploring intra-camera, inter-camera, and cross-modal domain information through a bottom-up strategy. DCCL [26] advanced this by proposing a dual consistency constraint framework combining camera proxies and class constraints, integrating label refinement with feature interaction learning for improved performance. However, these methods overlook the training instability caused by multi-objective optimization, a critical issue in UVI-ReID. SCA-RCP [53] groups cross-modal proxies of the same identity, it remains an extension of class-level cross-modal training, making it prone to errors from incorrect class mappings.

Our method divides contrastive learning into two phases. The first phase establishes robust intra-modal relationships through camera proxy-based contrastive learning, effectively addressing multi-objective training challenges with a novel camera-specific loss function that surpasses previous multi-loss approaches. The second phase integrates camera proxy information with class-level constraints, generating independent cross-modal mapping graphs at both levels. Furthermore, we resolve the cluster distribution-DBSCAN preset mismatch [28] in UVI-ReID by refining the memory bank update strategy.

C. Re-Ranking in Person Re-Identification

Re-ranking is a common post-processing method in image retrieval, generally divided into two approaches. Learning-based methods [54] incurred a significant amount of additional computation, making non-learning-based methods a more favorable choice. Recently, MCJA [55] proposed a multi-spectral enhancement ranking strategy to compute distances between multi-spectral embeddings during testing, but it could only operate in a supervised environment. Hu et al. [56] introduced a quantum-inspired interactive ranking aggregation (QI-IRA) method that uses quantum theory to model and aggregate multiple basic rankings, achieving performance close to or better than fully supervised methods with lower labeling costs. However, it still requires labeled data, making it unsuitable for unsupervised scenarios. In unsupervised settings, complex multi-peak relationships cause neighbor isolation issues. CMRR [57] addressed this with self-modal and cyclic-modal searching, but these methods increased computational overhead. CMKR [58] used intra-modal and inter-modal queries to tackle neighbor isolation, further raising computational costs. Zhang et al. [59] proposed a decentralized feature propagation algorithm for parallel or distributed computing, extended to cross-camera retrieval with improved feature propagation. However, incorporating camera information to this feature propagation will added extra time consumption.

To address these issues, we combine camera domain information with extracted features in the post-processing stage, replacing the simple k-nearest neighbor graph. This creates a more robust basis for graph propagation and avoids the time cost of integrating camera information during propagation. The method can be GPU-accelerated, significantly improving performance with minimal added computational time.

III. The Proposed Methods

In this section, we build upon the baseline framework to introduce our proposed two-stage training network, CEIL. The training process is shown in Fig.3, which comprises three components: the DCMU strategy, the CDL loss, and the PCM module. In the first stage, we exclusively employ camera proxy contrastive learning. In the second stage, PCM is integrated to enable cross-modal feature learning at both the camera proxy and class levels. Throughout the entire process, we use DCMU as the memory update strategy and apply CDL loss as the loss function for the camera proxy. After training, the GCR method is introduced to further optimize the results.

A. Baseline

As a pioneering pseudo-label-based method in the UVI-ReID field, ADCA [24] serves as our baseline. To ensure fair comparison with our CEIL, we exclude the cross-modal memory aggregation while incorporating Multikernel Maximum Mean Discrepancy (MMD) [60] loss into the objective function. Unlike DG-VAE [61], which employs distinct feature extractors and encoders for different modalities before VAE decoding, ADCA demonstrates the redundancy of such architecture in pseudo-label-based contrastive learning. It only utilizes modality-specific first layers in the pre-trained network to mitigate cross-modal discrepancies.

At the beginning of each epoch, features are extracted from both infrared and visible modal samples using the backbone. Subsequently, the DBSCAN [62] is applied for clustering these features. Following the clustering process, the memory for each cluster is initialized by computing the average feature values within each respective cluster, as described below:

$$m_q^e = \frac{1}{\left|\mathcal{C}_q^e\right|} \sum_{s \in \mathcal{C}_q^e} f^s, \tag{1}$$

where $e = \{v, r\}$ denote the visible and infrared modalities, respectively. $\mathcal{C}_q^e$ represents the q-th cluster set under modality e, $|\cdot|$ denotes the number of instances within each cluster, and f^s represents the features extracted from the image s.

During training, the modality-specific memories are updated through a momentum update strategy, as follows:

$$m_q^{e(\delta)} \leftarrow \beta m_q^{e(\delta-1)} + (1-\beta) f_q, \tag{2}$$

where f_q represents the features obtained from the training batch corresponding to m_q, β denotes the momentum updating factor, and δ signifies the iteration number. The loss for each modality is calculated using the InfoNCE loss [63], as follows:

$$\mathcal{L}_{\text{InfoNCE}}^e = -\log \frac{\exp\left(q^e \cdot m_+^e/\tau\right)}{\sum_{p=0}^{N^e} \exp\left(q^e \cdot m_q^e/\tau\right)}, \tag{3}$$

where m_+^e is the memory corresponding to the pseudo label of the query q^e, τ is a temperature parameter, and N^e is the

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number of clusters for modality e. The MMD loss, using a Gaussian kernel, aligns the feature distributions:

$$\mathcal{L}_{\text{MMD}} = \left\| \mathbb{E}_v \left[\phi \left(\mathbb{F}_N^v \right) \right] - \mathbb{E}_r \left[\phi \left(\mathbb{F}_N^r \right) \right] \right\|_{\mathcal{H}_k}^2, \tag{4}$$

where $\phi(\cdot)$ is a feature mapping function, and $\mathcal{H}_k$ represents the Reproducing Kernel Hilbert Space. The image features are denoted by $\mathbb{F}_N^v$ and $\mathbb{F}_N^r$. Finally, the total loss combines the proxy losses of each modality and MMD losses:

$$\mathcal{L}_{\text{base}} = \mathcal{L}_{\text{InfoNCE}}^v + \mathcal{L}_{\text{InfoNCE}}^r + \mathcal{L}_{\text{MMD}}. \tag{5}$$

B. Camera Domain-Driven Local Association Loss

The contrastive loss, using one-hot pseudo-labels [35], fails to accurately adjust relationships between camera proxies within the same class and those in other classes. PPCL [11] incorporates camera information to enhance the learning process. For each training sample, it mines credible proxies from camera agents that do not belong to the same class and are captured by non-self cameras. However, in cross-modal scenarios, the significant differences within and between modalities can lead to unstable clustering. When the clustering within the same camera is unreliable, attempting to perform credible mining across cameras becomes even more impractical. To address this limitation, we propose the CDL Loss ($\mathcal{L}_{cdl}$). Unlike PPCL, it conducts local contrastive learning within each camera domain, which avoids interference across camera domains while reducing the impact of unreliable clustering. Such a design is particularly well-suited for effectively handling feature learning in multi-camera scenarios.

For a given query feature, similarity is assessed with all camera proxies in the same class. By calculating the distance to both positive and negative proxies within each camera domain separately, reduces disturbances from domain differences and brings camera domains of the same class closer together. Given a query q, the loss $L(q, c)$ under camera c as follows:

$$L(q, c) = \frac{\exp(\text{Sim}^+(q, c))}{\exp(\text{Sim}^+(q, c)) + \sum_{i=1}^{n} \exp(\text{Sim}_i^-(q, c))}, \tag{6}$$

where $\text{Sim}^+(q, c)$ is the similarity between a query q and its positive proxy in camera domain c, $\text{Sim}_i^-(q, c)$ is the similarity between q and the i-th nearest negative proxy in this domain, and n is the number of nearest negative proxies considered per domain. The positive proxy is the proxy corresponding to the class of query q within the current camera domain. The nearest negative proxies are chosen based on the cosine similarity between query q and other proxies in the camera domain, with "nearest" meaning sorted in descending order of similarity. The overall loss $\mathcal{L}_{cdl}(q)$ is obtained by averaging over all effective camera domains as follows:

$$\mathcal{L}_{cdl}(q) = \frac{1}{|C_q|} \sum_{c \in C_q} -\log\left(L(q, c)\right), \tag{7}$$

where C_q denotes the set of valid camera domains for the query q, and $|C_q|$ represents their number.

Calculating losses based on local negative samples across different camera domains is crucial. Firstly, it helps maintain domain consistency. Images from different camera domains can vary significantly due to factors like shooting angles and lighting conditions. By calculating losses independently for each camera domain, the model can better handle these variations, improving its ability to generalize across different domains. Secondly, selecting negative samples with higher similarity to the current sample allows the model to learn more subtle differences. When negative samples are very similar to positive samples, the model is forced to learn more detailed features, enhancing its discriminative ability.

C. Difficulty-Aware Cluster-Based Memory Updating

Traditional momentum update methods facilitate cross-iteration information exchange by sequentially updating the corresponding centroids in the memory bank with fixed weights for each sample, However, which relies too much on randomness, often neglecting hard samples. A common issue is that, after multiple iterations, easy samples cluster well, but hard samples remain problematic. Shi et al. [51] selected a fixed number of hard samples from all clusters for updates, which performed poorly in the face of imbalanced cluster sizes. To address this limitation, we propose the DCMU strategy. It dynamically selects hard samples within aggregation clusters, generates weights based on difficulty and a smoothing factor, and then replaces the cluster centroids by cluster. As illustrated in Figure 4, this mechanism forces easy samples to converge towards hard samples, thereby significantly enhancing the model's ability to handle challenging samples.

During forward propagation, DCMU updates in units of clusters. For the i-th input feature vector $\mathbf{x}_i^c$ belonging to cluster c in the current input batch and the corresponding cluster center feature $\mathbf{f}^c$, the cosine distance is defined as:

$$d_i^c = 1 - \cos(\theta) = 1 - \frac{\mathbf{x}_i^c \cdot \mathbf{f}^c}{\|\mathbf{x}_i^c\| \|\mathbf{f}^c\|}, \tag{8}$$

where d_i^c is the cosine distance between $\mathbf{x}_i^c$ and $\mathbf{f}^c$, with $\|\cdot\|$ denotes the L2 norm. The hard samples for cluster c, denoted by S^c, are selected using the binary indicator F^c as follows:

$$\begin{aligned} F_i^c &= \begin{cases} 1 & \text{if } d_i \geq \mu^c \\ 0 & \text{otherwise} \end{cases}, \\ \mu^c &= \frac{1}{N^c} \sum_{i=1}^{N^c} d_i^c, \\ S^c &= \{s \mid F_s^c = 1\}, \end{aligned} \tag{9}$$

where μ^c represents the mean distance of all N^c samples in the current cluster. Then weights for hard samples w^c are calculated using a softmax and smoothed with a factor α:

$$\begin{aligned} w_j^c &= \text{Softmax}(d_j^c) \cdot \alpha + \delta_{\text{argmax}(d^c)}(j) \cdot (1 - \alpha), \\ \text{Softmax}(d_j^c) &= \frac{\exp(d_j^c)}{\sum_{s \in S^c} \exp(d_s^c)}, \end{aligned} \tag{10}$$

where j denotes the j-th index in S^c, and $\delta_{\text{argmax}(d^c)}$ is a binary list corresponding to d^c, with 1 assigned to the maximum value in d^c and 0 to all others. Finally, the pseudo-cluster

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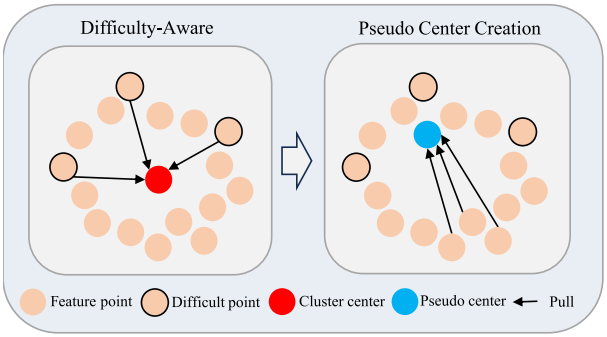

Fig. 4. Illustration of the purpose of DCMU. Cluster centers calculated by means and updated with momentum rely on randomness, leading to unfocused results. DCMU focuses on difficult samples and uses pseudo-centers to push nearby samples towards these challenging cases, resulting in tighter clusters.

center $\mathbf{f}^{pseudo^c}$ for cluster c is obtained by applying weighted aggregation to the selected features:

$$\mathbf{f}^{pseudo^c} = \frac{\sum_{s \in S^c} w_s^c \mathbf{x}_s^c}{\left\| \sum_{s \in S^c} w_s^c \mathbf{x}_s^c \right\|}. \tag{11}$$

During the back propagation phase, the gradient of the loss function $\mathcal{L}$ with respect to the input $\mathbf{x}_i^c$ is computed as:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{x}_i^c} = \frac{\partial \mathcal{L}}{\partial \mathbf{f}^c} \cdot \frac{\partial \mathbf{f}^c}{\partial \mathbf{x}_i^c} \approx \frac{\partial \mathcal{L}}{\partial \mathbf{f}^c} \cdot \mathbf{f}^c, \tag{12}$$

where $\frac{\partial \mathcal{L}}{\partial \mathbf{x}_i^c}$ refer to the gradient of the loss function $\mathcal{L}$ with respect to the input feature $\mathbf{x}_i^c$, can be obtained by considering both the direct gradient $\frac{\partial \mathcal{L}}{\partial \mathbf{f}^c}$ with respect to the cluster center feature $\mathbf{f}^c$ and the gradient of $\mathbf{f}^c$ with respect to $\mathbf{x}_i^c$, denoted as $\frac{\partial \mathbf{f}^c}{\partial \mathbf{x}_i^c}$, which represents $\mathbf{f}^c$. After completing the loss computation, the cluster centers are updated as follows:

$$\mathbf{f}^c \leftarrow \mathbf{f}^{pseudo^c}. \tag{13}$$

Updating cluster centers using pseudo-centroids achieves dual objectives. First, computing cosine distances between current samples and hard sample clusters from previous iterations accelerates the convergence of easy samples toward hard sample regions. Second, formed through weighted aggregations of historical hard instances, pseudo-centroids naturally lie closer to hard samples, thereby recalibrating the hardness levels of previously easy samples. This dynamic tug-of-war process drives the model to learn noise-resistant feature representations while reducing sensitivity to outliers.

D. Graph Network-Based Camera Constraint Adjustment Re-Ranking

Re-ranking techniques improve retrieval accuracy by refining initial results. In the UVI-ReID task, traditional re-ranking methods [29] can effectively enhance performance, but they often come with high computational costs. Graph Convolutional Network [64] (GNN) based methods, GCN [31], leverage GPU acceleration to significantly reduce computation time, yet they do not achieve further performance improvements. To balance low computational cost and high retrieval performance, we propose GCR. The core innovation of it compared to GCN lies in incorporating camera prior knowledge into the graph construction process and ensuring stability through multiple symmetric normalization operations during the iterative phase. Experimental results demonstrate that GCR achieves significant performance improvements without substantially increasing computation time. Detailed results are analyzed in the experiments section.

The original features of the combined query and gallery sets are denoted by X_u, with n images in total. We define a graph $G = (V, E)$ where each vertex v_i in $V = \{v_1, \ldots, v_n\}$ corresponds to an image, and the edges $E \in \mathbb{R}^{n \times n}$ represent the similarities between these images. Initially, cosine distances are calculated to establish a similarity ranking R for the images. Then, E is adjusted based on camera information:

$$E_{ij} = \begin{cases} \lambda, & \text{if } \text{cam}_i = \text{cam}_j \text{ and } j \in R_i \\ 1/\lambda, & \text{if } \text{cam}_i \neq \text{cam}_j \text{ and } j \in R_i \\ 0, & \text{otherwise} \end{cases}, \tag{14}$$

where sample j is a top k_1 nearest neighbor of sample i, with cam_i and cam_j as their respective cameras. λ adjusts weights, emphasizing inter-camera samples when smaller. The ranking R and the matrix S, which stores the squared similarity scores of the top k_2 neighbors, are saved for use in the second stage.

In the second phase, to enhance retrieval performance, feature aggregation is necessary. Traditional re-ranking methods use local query expansion [29], while GCR, like GCN [31], performs iterative propagation on a GNN. Unlike GCN, which only performs the conversion to an undirected graph and normalization after initialization, we execute this conversion process before each iteration, laying a solid foundation for matrix propagation, as shown below:

$$\tilde{E}^{(0)} = \text{diag}\left(\frac{1}{\|E + (E)^\top\|_1}\right) \cdot \left(E + (E)^\top\right), \tag{15}$$

where $E^{(0)}$ represents the adjacency matrix at the initial iteration, and its transpose is denoted by $(E^{(0)})^\top$. $\|\cdot\|_1$ denotes the L1 norm, which calculates the sum of the elements in each row of the matrix. $\text{diag}(\cdot)$ converts a vector into a diagonal matrix, ensuring that each element of the vector symmetrically scales the corresponding row of the matrix. This process results in the matrix $\tilde{E}^{(0)}$, which is both symmetrized and normalized, ensuring uniformity and balance in node relationships. The neighbor aggregation process for updating the adjacency matrix to $E^{(t)}$ at the t-th iteration is as follows:

$$E^{(t)}[i, j] = \sum_{m=0}^{k_2-1} S[i, m] \cdot \tilde{E}^{(t-1)}\big[R(i, m), j\big], \tag{16}$$

where $E^{(t)}[i, j]$ denotes the updated value between nodes i and j. The aggregation sums over neighboring nodes, where $S[i, m]$ is the similarity weight between node i and its m-th neighbor. $R(i, m)$ gives the index of the m-th neighbor of node i. $\tilde{E}^{(t-1)}[r(i, m), j]$ retrieves the symmetrized and normalized adjacency value from the previous iteration. To facilitate understanding, the complete process of post-processing using GCR is provided in Algorithm 1.

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Algorithm 1 Re-Ranking Process of the Proposed GCR

Input: The union of query and gallery features X_u and their corresponding camera domains C_u.
Parameter: Hyper-parameters k_1, k_2, and T.
Output: Final distance matrix E.

1: Calculate the similarity matrix S and similarity ranking R using X_u, k_1 and k_2.
2: Compute the adjacency matrix E using R and C_u according to Eq.(14).
3: Initialize $t = 0$.
4: **while** $t < T$ **do**
5: Translate the adjacency matrix E into an undirected graph and perform row normalization, by using Eq.(15).
6: Update matrix E through message propagation with GNN using S according to Eq.(16).
7: Increment t
8: **end while**
9: **return** E.

E. Proxy-Enhanced Cross-Modal Mapping

Using only MMD for aligning feature distributions often results in identity misalignment between visible and infrared modalities. Previous methods, like PGM [27], address this by clustering at the class level and establishing mappings between modalities, but they overlook camera proxy information. To overcome this, we introduce the PCM module in the second training stage. Building on the foundation of relatively stable and reliable modal cohesive classes, this module constructs cross-modal correspondences at both the camera proxy and class levels, significantly enhancing robustness.

First, we construct independent bipartite graph structures for the proxy clusters and identity clusters obtained from the infrared and visible modalities. This allows each proxy cluster and identity cluster within the infrared modality to find a corresponding cluster within the visible modality. For the proxy clusters and identity clusters that do not form pairs in the initial matching, we repeat the above process until all clusters have corresponding matches. In class-based matching, PGM has demonstrated that cross-modal bipartite graph matching is essentially a binary linear programming problem with linear constraints. The matching cost penalizes feature differences between matched clusters across modalities, meaning that more similar cluster features result in lower costs. The proposed PCM, which introduces additional cross-modal camera proxies, also adheres to this principle. Following PGM, we utilize the Hungarian algorithm [65] for matching, and maintain its cost matrix as follows:

$$Cost(i, j) = -\frac{1}{\exp(Dis(i, j))}, \tag{17}$$

where $Cost(i, j)$ represents the cost function between the proxies or classes of i and j, and $Dis(i, j)$ denotes the distance measure between them, obtained from our GCR, which takes additional consideration of camera domain information, which can provide a more comprehensive measure. In Fig.5 provides a simplified diagram of the mapping process.

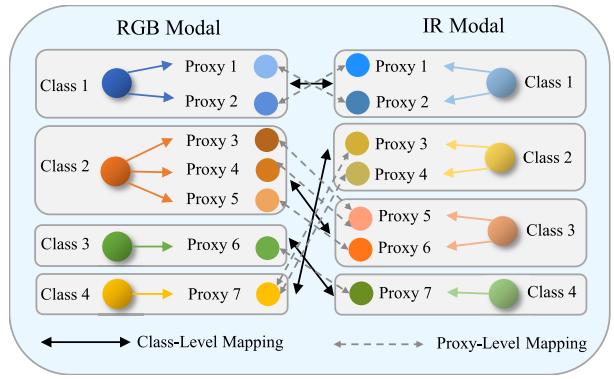

Fig. 5. Illustration of cross-modal mapping in PCM. PCM creates cross-modal mappings between classes and camera proxies, significantly boosting error robustness through leveraging consistency between these elements.

Secondly, after obtaining the mapping relationships T^p_{i2r}, T^p_{r2i}, T^c_{i2r}, and T^c_{r2i} between the two domains regarding the proxy clusters and class clusters, using these mapping relationships to perform modal transformation on the pseudo labels P_v, P_r, C_v, and C_r obtained through clustering:

$$\begin{aligned} P_{v2r} &= T^p_{i2r}(P_v), \quad P_{r2v} = T^p_{r2i}(P_r), \\ C_{v2r} &= T^c_{r2i}(C_v), \quad C_{r2v} = T^c_{r2i}(C_r), \end{aligned} \tag{18}$$

during stage 2, we not only use the pseudo-labels generated by clustering to train within the same modality but also utilize the generated mapping pseudo-labels to train across modalities. Since the incorporation of the class hierarchy, the calculation of the loss differs from stage 1. To begin with, we compute the intra-modal loss, which is composed of two distinct components: The proxy-level loss is performed using our proposed $\mathcal{L}^{\text{intra}}_{cdl}$, which utilizes P_v and P_r, while the class-level loss is accomplished through $\mathcal{L}^{\text{intra}}_{\text{InfoNCE}}$, which takes C_v and C_r as inputs, as shown below:

$$\mathcal{L}^{\text{intra}} = \mathcal{L}^{\text{intra}}_{cdl} + \mathcal{L}^{\text{intra}}_{\text{InfoNCE}}, \tag{19}$$

this combination of losses at different levels helps our model learn better. The calculation for cross-modal loss and intra-modal loss is essentially symmetrical, which ensures that the updates of the infrared and visible light modes are balanced during training. The difference lies in replacing the class-level labels with C_{v2r} and C_{r2v}, and the proxy-level labels with P_{v2r} and P_{r2v}. Algorithm 2 illustrates the operations of our proposed PCM during the second phase of CEIL.

F. Unified Loss

We use $\mathcal{L}_{\text{cdl}}$ and $\mathcal{L}_{\text{MMD}}$ to define the first stage loss as:

$$\mathcal{L}_{\text{stage1}} = \mathcal{L}^v_{\text{cdl}} + \mathcal{L}^r_{\text{cdl}} + \mathcal{L}_{\text{MMD}}, \tag{20}$$

where the superscript of the $\mathcal{L}_{\text{cdl}}$ indicating its modality. And define the loss of the second stage as follows:

$$\mathcal{L}_{\text{stage2}} = \mathcal{L}^{\text{intra}} + \mathcal{L}^{\text{cross}}, \tag{21}$$

where $\mathcal{L}_{intra}$ and $\mathcal{L}_{cross}$ are incorporated into the computations for both the visible and infrared modalities.

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Algorithm 2 Training Process for CEIL: Stage 2

Input: Unlabeled image $I^v = \{x_i^v\}_{i=1}^{N_v}$ and $I^r = \{x_i^r\}_{i=1}^{N_r}$, feature extractor E trained after stage 1, memory bank M.
Parameter: Training epochs and iters.
Output: Feature extractor E trained post-stage 2.

```
for i = 1 to epochs do
    Use E to extract features from I^v and I^r, obtain the feature sets F^v and F^r.
    Apply DBSCAN to cluster the features F^v and F^r for each modality C_v and C_r.
    Partition the camera proxies into P_v and P_r according to their respective camera sources.
    Obtain cross-modal mappings T^p_{i2r}, T^p_{r2i}, T^c_{i2r}, and T^c_{r2i} between camera proxies and class proxies by PCM.
    Obtain cross-modal mapping labels P_{v2r}, P_{r2v}, C_{v2r}, and C_{r2v} according to Eq.(18).
    for j = 1 to iters do
        Update M based on DCMU.
        Using P^v, P^r, C_v and C_r, calculate the intra-modal loss L^intra according to Eq.(19).
        Similarly, using P_{v2r}, P_{r2v}, C_{v2r}, and C_{r2v}, calculate the cross-modal loss L^cross.
        Optimize E to minimize the loss defined in Eq.(21).
    end for
end for
return E.
```

IV. Experiments

A. Experimental Setting

1) Datasets: We evaluate our CEIL using three public ReID datasets: SYSU-MM01 [14], RegDB [13], and LLCM [15].

The SYSU-MM01 dataset includes 30,071 visible images and 15,792 near-infrared images captured by 4 visible and 2 infrared cameras. The dataset is divided into a training set with 395 identities and a test set with 96 identities. There are two matching modes: "All-search", which matches images across all cameras, and "Indoor-search", which restricts matches to indoor cameras only. In both modes, infrared images are matched with visible images. During testing, one image per identity per camera is randomly selected to form a gallery set for evaluation. This process is repeated 10 times with different random gallery splits, and the average performance is reported.

The RegDB dataset comprises 412 identities, each with 10 visible images and 10 infrared images. According to the protocol by Ye et al. [66], the dataset is evenly split, with half of the identities used for training and the other for testing. There are two matching modes: "VIS to IR" for matching visible to infrared images, and "IR to VIS" for the reverse.

The LLCM dataset, collected under low-light conditions, includes a training set with 713 identities (16,946 visible images and 13,975 infrared images) and a test set with 351 identities (8,680 visible images and 7,166 infrared images). Similar to the RegDB dataset, both "VIS to IR" and "IR to VIS" modes are employed to evaluate the model's performance. The selection of samples during the evaluation process is similar to that used for the SYSU-MM01 dataset.

2) Evaluation Metrics: We use Cumulative Matching Characteristics (CMC), Mean Average Precision (mAP), and Mean Inverse Negative Penalty (mINP) [5] as the main evaluation metrics to validate the effectiveness of our proposed CEIL.

3) Implementation Details: Our model uses the same feature extractor as CCLNet [42]. Specifically, this feature extractor is a dual-stream ResNet that has been pre-trained on ImageNet [67]. All images are resized to 288 × 144 and using data augmentation such as horizontal flipping, random erasing, and channel augmentation [16], following the paper of PGM [27]. Based on the magnitude of the data, we have selected different types of training minibatch. For the SYSU-MM01 and LLCM datasets, we selected 8 identities, each with 10 visible light images and 10 infrared images. For the RegDB dataset, both the number of identities per modality and the corresponding number of images are halved. We use warm-up to update the learning rate from 3.5e-6 to 3.5e-4 during the first 10 epochs. At the 20th and 50th epochs, the learning rates are adjusted to 3.5e-5 and 3.5e-6. We employ the Adam optimizer [68] to train for a total of 90 epochs, entering the second stage after the 50st epoch. The implementation is carried out in PyTorch and runs on an NVIDIA RTX 3090 GPU.

B. Comparison With SOTA Methods

The results for the SYSU-MM01 and RegDB datasets are shown in Table I. Meanwhile, the results for the LLCM dataset are presented in Table II. Among these, CCLNet, PGM, and SDCL were reproduced based on local experiments due to the availability of comprehensive code provided for these methods. In both tables, the symbol "*" indicates methods that have been post-processed using GCR. The sections above the horizontal line highlight the performance of supervised learning methods, whereas the sections below showcase our CEIL method along with other unsupervised learning approaches.

1) Comparison With VI-ReID Methods: It's encouraging that our CEIL, even without post-processing, matches the performance of supervised VI-ReID methods like AGW and PMT. When enhanced with GCR, it significantly outperforms top supervised methods on the SYSU-MM01 and RegDB datasets. The LLCM dataset, which presents challenges such as low light and climate changes, which difficult for unsupervised learning. However, with the addition of GCR, CEIL remains competitive with SOTA models like HOS-Net. Although there is a noticeable performance gap between CEIL without GCR and the current state-of-the-art supervised methods on more demanding datasets, the significant improvement brought about by post-processing highlights CEIL's potential and effectiveness. This also emphasizes the critical role of GCR, which helps CEIL catch up with minimal extra overhead.

2) Comparison With UVI-ReID Methods: Even without our proposed GCR, the CEIL model we developed significantly outperforms the current state-of-the-art CMAM model on the SYSU-MM01, RegDB, and LLCM datasets, achieving a comprehensive lead. This clearly highlights the effectiveness of leveraging readily accessible camera information. Obtaining

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TABLE I

COMPARISONS WITH SOTA METHODS ON THE SYSU-MM01 AND REGDB DATASET. FOR THE SYSU-MM01 DATASET, THE ANALYSIS INCLUDES ALL SEARCH AND INDOOR SEARCH MODES, WHILE FOR THE REGDB DATASET, IT COVERS VIS TO IR AND IR TO VIS MODES. ABOVE THE HORIZONTAL LINE IS THE VI-REID METHOD, BELOW IS THE UVI-REID METHOD WHERE THE BOLDED ONE PERFORMS THE BEST. IN ALL TABLES, "∗" DENOTES GCR POST-PROCESSING, AND "†" INDICATES THE METHOD USING CAMERA PROXY. UNMARKED DATA ARE PERCENTAGES UNLESS SPECIFIED OTHERWISE

Methods		SYSU-MM01		RegDB	
		All search	Indoor search	VIS to IR	IR to VIS
Name	Reference	R-1 / mAP / mINP	R-1 / mAP / mINP	R-1 / mAP / mINP	R-1 / mAP / mINP
CA [16]	ICCV 2021	69.9 / 66.9 / 53.6	76.3 / 80.4 / 76.8	85.0 / 79.1 / 65.3	84.8 / 77.8 / 61.6
AGW [5]	TPAMI 2021	47.5 / 47.7 / 35.3	54.2 / 63.0 / 59.2	70.1 / 66.4 / 50.2	70.5 / 65.9 / 51.2
MAUM [17]	CVPR 2022	71.7 / 68.8 / -	77.0 / 82.0 / -	87.9 / 85.1 / -	87.0 / 84.3 / -
MSCLNet [18]	ECCV 2022	77.0 / 71.6 / -	78.5 / 81.2 / -	84.2 / 81.0 / -	83.9 / 78.3 / -
PartMix [19]	CVPR 2023	77.8 / 74.6 / -	81.5 / 84.4 / -	85.7 / 82.3 / -	85.0 / 82.5 / -
PMT [20]	AAAI 2023	67.5 / 65.0 / 51.9	71.7 / 76.5 / 72.7	84.8 / 76.6 / -	84.2 / 75.1 / -
MSALNet [69]	TCSVT 2024	71.2 / 68.0 / 54.6	79.4 / 82.8 / 79.3	90.2 / 84.3 / 70.5	91.1 / 85.2 / 72.7
DMA [22]	TIFS 2024	74.6 / 70.4 / -	82.9 / 85.1 / -	93.3 / 88.3 / -	91.5 / 86.8 / -
CycleTrans [21]	TNNLS 2024	76.6 / 72.6 / -	87.2 / 85.0 / -	90.6 / 85.6 / -	91.5 / 87.0 / -
HOS-Net [23]	AAAI 2024	75.6 / 74.2 / -	84.2 / 86.7 / -	94.7 / 90.4 / -	93.3 / 89.2 / -
OTLA [41]	ICCV 2021	29.9 / 27.1 / -	29.8 / 38.8 / -	32.9 / 29.7 / -	32.1 / 28.6 / -
ADCA [24]	MM 2022	45.5 / 42.7 / 28.9	50.6 / 62.1 / 55.2	67.4 / 65.6 / 52.7	69.5 / 63.8 / 49.6
CCLNet [42]	MM 2023	55.3 / 50.4 / 33.9	56.5 / 63.7 / 59.6	72.9 / 71.5 / 64.9	72.2 / 70.4 / 62.0
PGM [27]	CVPR 2023	56.3 / 51.4 / 35.1	58.0 / 64.9 / 60.7	70.3 / 66.4 / 54.2	70.8 / 65.5 / 51.0
CHCR [46]	TCSVT 2024	47.7 / 45.3 / -	50.1 / 42.2 / -	68.2 / 63.8 / -	69.1 / 64.0 / -
SDCL [70]	CVPR 2024	61.3 / 59.8 / 46.5	69.2 / 75.1 / 71.4	86.9 / 78.9 / 62.8	85.8 / 77.3 / 59.6
CMAM [48]	TCSVT 2024	62.0 / 58.2 / -	67.6 / 74.0 / -	89.1 / 74.0 / -	89.0 / 74.0 / -
RPNR [45]	MM 2024	65.2 / 60.0 / -	68.9 / 74.4 / -	90.9 / 84.7 / -	90.1 / 83.2 / -
PCAL [49]	TIFS 2025	58.0 / 52.9 / 36.9	60.1 / 66.7 / 62.1	86.4 / 82.5 / 72.3	86.2 / 81.2 / 68.7
MMM [44]	ECCV 2025	65.9 / 61.8 / -	70.3 / 74.9 / -	89.6 / 83.7 / -	87.0 / 80.9 / -
GUR† [52]	ICCV 2023	63.5 / 61.6 / 47.9	71.1 / 76.2 / 72.6	73.9 / 70.2 / 58.8	75.0 / 69.9 / 56.2
SCA-RCP† [53]	TKDE 2024	51.4 / 48.5 / 33.6	56.8 / 64.2 / 59.3	85.6 / 79.1 / -	82.4 / 75.7 / -
DCCL† [26]	TIFS 2024	63.2 / 58.6 / 43.0	66.7 / 71.8 / 67.5	78.3 / 71.3 / 58.8	78.2 / 71.3 / 55.2
CEIL†	Ours	68.2 / 63.7 / 48.7	73.3 / 77.0 / 72.9	92.0 / 89.1 / 82.8	92.6 / 89.1 / 81.8
CEIL†∗	Ours	**86.8 / 84.0 / 75.2**	**91.3 / 92.2 / 90.6**	**94.7 / 95.5 / 95.5**	**95.2 / 95.7 / 95.4**

TABLE II

COMPARISONS WITH SOTA METHODS ON THE LLDM DATASET, INCLUDING VIS TO IR AND IR TO VIS MODES

Methods		VIS to IR	IR to VIS
Name	Reference	R-1 / mAP	R-1 / mAP
CA [16]	ICCV 2021	56.5 / 59.8	48.8 / 56.6
AGW [5]	TPAMI 2021	51.5 / 55.3	43.6 / 51.8
DART [71]	CVPR 2022	60.4 / 63.2	52.2 / 59.8
DEEN [15]	CVPR 2023	62.5 / 65.8	54.9 / 62.9
HOS-Net [23]	AAAI 2024	64.9 / 67.9	56.4 / 63.2
OTLA [41]	ICCV 2021	17.8 / 20.4	14.9 / 18.6
ADCA [24]	MM 2022	23.5 / 28.2	16.1 / 21.4
CCLNet [42]	MM 2023	45.3 / 49.9	39.3 / 45.3
PGM [27]	CVPR 2023	44.4 / 48.6	38.4 / 44.2
SDCL [70]	CVPR 2024	46.9 / 52.4	43.4 / 48.2
SCA-RCP† [53]	TKDE 2024	29.1 / 33.3	22.3 / 28.0
CEIL†	Ours	51.6 / 55.1	44.4 / 50.0
CEIL†∗	Ours	**65.9 / 68.2**	**58.0 / 62.7**

this information is straightforward, as it is often embedded in image metadata or easily available from recording devices. Given this ease of access, utilizing it becomes a natural and logical step. With the application of our GCR technique, the performance gap widens even further. When an additional post-processing step can deliver such substantial performance improvements without adding significant computational overhead, there is no reason not to employ it. Notably, after applying GCR, we achieved over 60% mAP on the particularly challenging LLCM dataset and exceeded 80% mAP on the less challenging SYSU-MM01 dataset. This indicates that the performance is nearly ready for practical application and deployment in real-world scenarios.

C. Ablation Studies

We conducted ablation studies on the RegDB, LLCM, and SYSU-MM01 datasets to demonstrate the effectiveness of each component in the training of the CEIL model. The ablation experiments on RegDB and SYSU-MM01 are presented in Table III, while the experiments on LLCM are shown in Table IV. In these tables, "Baseline" refers to the baseline model that does not incorporate any of our proposed methods. In Table III and Table IV, "Time" denotes training time. In Table VII, it refers to post-processing time during testing.

1) Effectiveness of PCM: The comparative analysis of Strategy 2 versus Strategy 4, and Strategy 3 versus Strategy 5, substantiates the efficacy of the PCM module in enhancing performance, highlighting its effectiveness in bridging the modality gap. To corroborate the utility of camera proxy mapping, an array of cross-modality correspondence methodologies were trialed in the secondary training phase on the SYSU-MM01 dataset, as delineated in Table V. The experimental results demonstrate that combining camera proxies with class matching effectively reduces errors and minimizes the risk of misguidance caused by class matching inaccuracies.

2) Effectiveness of DCMU: The comparisons between Baseline versus Strategy 1, Strategy 2 versus Strategy 3, and

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TABLE III

THE INFLUENCE OF EACH COMPONENT ON THE PERFORMANCE OF THE CEIL ON SYSU-MM01 AND REGDB. "PCM" DENOTES PROXY-ENHANCED CROSS-MODAL MAPPING. "DCMU" DENOTES DIFFICULTY-AWARE CLUSTER-BASED MEMORY UPDATING. "$\mathcal{L}_{cdl}$" DENOTES CAMERA DOMAIN-DRIVEN LOCAL ASSOCIATION LOSS. "GCR" DENOTES GRAPH NETWORK-BASED CAMERA CONSTRAINT ADJUSTMENT RE-RANKING

Methods	Settings				SYSU-MM01			RegDB		
					All search	Indoor search	Time (s)	VIS to IR	IR to VIS	Time (s)
	PCM	DCMU	$\mathcal{L}_{cdl}$	GCR	R-1 / mAP / mINP	R-1 / mAP / mINP		R-1 / mAP / mINP	R-1 / mAP / mINP	
Baseline					45.0 / 42.3 / 27.7	49.9 / 57.4 / 52.3	8.706 x 10^3	35.8 / 35.7 / 28.4	38.7 / 38.5 / 30.2	1.847 x 10^3
Strategy 1		✓			48.5 / 46.5 / 32.1	53.9 / 60.6 / 55.8	8.570 x 10^3	49.8 / 47.6 / 36.5	51.8 / 48.9 / 37.1	1.832 x 10^3
Strategy 2			✓		48.1 / 43.8 / 28.4	53.4 / 61.7 / 57.5	8.523 x 10^3	42.8 / 41.4 / 31.5	43.6 / 41.9 / 31.6	1.877 x 10^3
Strategy 3		✓	✓		52.9 / 49.2 / 33.8	59.1 / 64.9 / 60.1	8.474 x 10^3	53.7 / 50.2 / 37.6	54.2 / 48.8 / 35.4	1.858 x 10^3
Strategy 4	✓		✓		60.5 / 56.5 / 41.3	64.1 / 69.8 / 65.1	1.495 x 10^4	66.2 / 65.6 / 58.5	66.4 / 65.5 / 58.2	3.254 x 10^3
Strategy 5	✓	✓	✓		68.2 / 63.7 / 48.7	73.3 / 77.0 / 72.9	1.476 x 10^4	92.0 / 89.1 / 82.8	92.6 / 89.1 / 81.8	3.208 x 10^3
Strategy 1*		✓		✓	69.9 / 69.0 / 60.3	75.6 / 79.9 / 78.1	-	36.5 / 36.5 / 37.5	39.4 / 39.9 / 38.5	-
Strategy 2*			✓	✓	65.7 / 65.0 / 56.0	71.4 / 76.6 / 74.4	-	43.2 / 45.8 / 47.8	44.5 / 46.7 / 48.5	-
Strategy 3*		✓	✓	✓	72.9 / 71.8 / 63.4	78.8 / 82.2 / 80.4	-	54.4 / 58.3 / 59.8	55.4 / 57.1 / 58.3	-
Strategy 4*	✓		✓	✓	79.8 / 78.4 / 69.4	84.5 / 87.1 / 85.2	-	68.7 / 72.9 / 74.4	69.4 / 73.2 / 74.3	-
Strategy 5*	✓	✓	✓	✓	**86.8 / 84.0 / 75.2**	**91.3 / 92.2 / 90.6**	-	**94.7 / 95.5 / 95.5**	**95.2 / 95.7 / 95.4**	-

TABLE IV

THE INFLUENCE OF EACH COMPONENT ON THE PERFORMANCE OF THE PROPOSED CEIL ON LLCM

Methods	VIS to IR	IR to VIS	Time (s)
	R-1 / mAP / mINP	R-1 / mAP / mINP	
Baseline	35.8 / 40.0 / 34.6	33.5 / 38.7 / 34.8	1.028 x 10^4
Strategy 1	42.2 / 46.5 / 41.0	38.4 / 44.1 / 40.2	1.014 x 10^4
Strategy 2	36.4 / 40.5 / 34.8	34.0 / 39.6 / 35.9	1.009 x 10^4
Strategy 3	44.0 / 48.0 / 42.3	39.9 / 45.2 / 41.3	1.001 x 10^4
Strategy 4	45.5 / 49.8 / 44.0	40.9 / 46.6 / 42.6	1.684 x 10^4
Strategy 5	51.2 / 54.8 / 50.0	49.5 / 54.0 / 50.5	1.675 x 10^4
Strategy 1*	56.8 / 60.3 / 56.2	50.4 / 55.2 / 52.0	-
Strategy 2*	51.8 / 55.3 / 50.8	45.4 / 50.3 / 47.0	-
Strategy 3*	58.1 / 60.9 / 56.8	51.2 / 56.0 / 52.9	-
Strategy 4*	59.8 / 63.0 / 58.2	53.0 / 58.1 / 55.0	-
Strategy 5*	**65.9 / 68.2 / 63.8**	**58.0 / 62.7 / 59.6**	-

TABLE V

COMPARISON OF CROSS-MODAL TRAINING METHODS ON THE SYSU-MM01 DATASET IN BOTH ALL SEARCH AND INDOOR SEARCH MODES. "STAGE 1" UTILIZES ONLY MMD LOSS, "ONLY PROXY" FOCUSES ON CAMERA PROXY-LEVEL TRAINING, "ONLY CLASS" EMPHASIZES CLASS-LEVEL TRAINING, AND "PCM" DEMONSTRATES THE PERFORMANCE OF OUR PROPOSED METHOD

Methods	All search	Indoor search
	R-1 / mAP	R-1 / mAP
Stage 1	52.9 / 49.2	59.1 / 64.9
Only proxy	64.4 / 60.4	69.0 / 73.7
Only class	65.0 / 61.6	70.2 / 74.7
PCM (Ours)	**68.2 / 63.7**	**73.3 / 77.0**

TABLE VI

COMPARISON OF POST-PROCESSING TECHNIQUES APPLIED TO EXISTING METHODS ON THE SYSU-MM01 DATASET FOR ALL-SEARCH AND INDOOR-SEARCH. THE EVALUATED METHODS COMPRISE NONE WHICH MEANS NO POST-PROCESSING, KR, ECN, GCN, AND PROPOSED GCR

Methods	Post-Processing	All search	Indoor search
		R-1 / mAP	R-1 / mAP
CCLNet [42]	None	55.3 / 50.4	56.5 / 63.7
	KR [29]	68.2 / 67.5	72.3 / 77.6
	ECN [30]	67.9 / 67.1	73.2 / 78.1
	GCN [31]	68.4 / 67.7	73.6 / 78.8
	GCR (Ours)	**70.0 / 69.0**	**75.0 / 79.9**
PGM [27]	None	56.3 / 51.4	58.0 / 64.9
	KR	67.2 / 63.8	69.9 / 73.8
	ECN	67.9 / 66.1	69.7 / 74.2
	GCN	68.3 / 66.4	70.1 / 74.9
	GCR (Ours)	**69.8 / 67.7**	**71.4 / 75.8**
SDCL [70]	None	61.3 / 59.8	69.2 / 75.1
	KR	74.9 / 73.6	81.8 / 85.2
	ECN	75.5 / 76.2	81.8 / 85.3
	GCN	75.9 / 76.5	82.2 / 86.1
	GCR (Ours)	**77.5 / 78.0**	**83.5 / 87.0**

Strategy 4 versus Strategy 5 demonstrate that replacing the traditional momentum update strategy with DCMU, whether used independently or in combination with our proposed modules, leads to significant performance improvements across all datasets. The robust capability of DCMU in identifying challenging samples during training narrows the gap between hard and easy samples, resulting in more compact clustering.

3) Effectiveness of CDL Loss: The comparison between Baseline and Strategy 2 shows that adding the CDL loss alone improves performance. On the RegDB dataset, intra-modality samples are captured by a single camera and thus lack the benefits of camera information. Despite this, the CDL loss still surpasses the InfoNCE loss, with the improvement attributed solely to its ability to perform local smoothness computation.

4) Effectiveness of GCR During Test: GCR post-processing consistently enhances model performance across all training strategies, as shown by the improved results from Strategies 1 to 5 compared to the original Strategies 1∗ to 5∗. As a plug-and-play re-ranking module, GCR is highly versatile in existing methods. Table VI shows consistent performance enhancements from this post-processing. The experiment using features trained with CEIL on the SYSU-MM01 dataset for post-processing validated the effectiveness of GCR by comparing it with other unsupervised non-learning methods. As shown in Table VII. Like GCN [31], GCR uses GPU acceleration to significantly reduce computational overhead. Incorporating camera information, GCR builds a more reliable initial graph, boosting performance by over 2% with less than 10 milliseconds of additional computation. We conducted

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TABLE VII

COMPARISON OF OTHER RE-RANKING METHODS WITH THE PROPOSED GCR, INCLUDING PERFORMANCE AND COMPUTATIONAL TIME. ALL COMPARISONS ARE BASED ON PROPOSED CEIL

Methods	All search	Indoor search
	R-1 / mAP / Time (ms)	R-1 / mAP / Time (ms)
None	68.2 / 63.7 / -	73.3 / 77.0 / -
KR	83.9 / 79.3 / 2.96 x 10^4	87.5 / 88.2 / 2.87 x 10^4
ECN	83.3 / 81.6 / 1.99 x 10^4	87.4 / 89.5 / 1.67 x 10^4
GCN	83.7 / 82.5 / 9.72 x 10^1	88.3 / 90.0 / 6.51 x 10^1
GCR (Ours)	**86.8 / 84.0 / 1.08 x 10^2**	**91.3 / 92.2 / 6.71 x 10^1**

TABLE VIII

ABLATION STUDY OF "GCR" COMPONENTS. "MSN" REFERS TO THE APPLICATION OF SYMMETRIZATION AND NORMALIZATION AT EACH FEATURE AGGREGATION STEP. "CAMERA" REFERS TO ENHANCING THE INITIAL MATRIX USING CAMERA INFORMATION

Settings		All search	Indoor search
MSN	Camera	R-1 / mAP / Time (ms)	R-1 / mAP / Time (ms)
		84.4 / 81.3 / 8.83 x 10^1	89.1 / 90.4 / 5.51 x 10^1
✓		86.1 / 83.2 / 9.56 x 10^1	90.6 / 91.5 / 6.31 x 10^1
	✓	85.3 / 82.2 / 1.01 x 10^2	89.8 / 91.2 / 6.18 x 10^1
✓	✓	**86.8 / 84.0 / 1.08 x 10^2**	**91.3 / 92.2 / 6.71 x 10^1**

ablation studies on the components of GCR as shown in Table VIII, where "MSN" indicates symmetric normalization before each feature aggregation, and "Camera" represents the incorporation of camera information. The results highlight the positive impact of each component on GCR.

5) Cost Analysis: During training, CDL Loss transforms class domains into camera domains for local smoothing, replacing the traditional InfoNCE Loss. As demonstrated in the comparison between Baseline and Strategy 3, this replacement reduces training time and improves performance on the SYSU-MM01 and LLCM datasets. However, for single-camera domains such as RegDB, it introduces a slight overhead of approximately 30 seconds. Compared to traditional momentum updates, DCMU creates pseudo-centroids during forward propagation but reduces backpropagation costs, ultimately saving time, as shown in the comparison between Baseline and Strategy 1. PCM introduces a second training stage, increasing time by 60%, as seen in the comparison between Strategy 3 and Strategy 5, but it delivers significant performance improvements, particularly on RegDB, where gains exceed 60%, making it a worthwhile trade-off.

During testing, GCR based on cosine distance constructs an initial matrix and performs aggregation propagation by utilizing camera information. Both the time complexity and space complexity of this method are at the $O(N^2)$ level, which is consistent with the complexity of cosine similarity calculations. Specifically, the primary bottleneck in time complexity lies in the GNN propagation process and the computation of initial similarity, while the space complexity is mainly used to store the adjacency matrix and initial similarity matrix required for iterations, where N represents the number of test samples. Despite facing the same complexity challenges, our proposed GCR significantly outperforms traditional cosine distance methods in terms of performance.

TABLE IX

ILLUSTRATION OF HYPERPARAMETERS: FOR THE DCMU, $\mathcal{L}_{cdl}$, AND GCR. THE SUPERSCRIPTS INDICATE USAGE PHASES

Datasets	$\text{DCMU}^{\text{train}}$	$\mathcal{L}_{cdl}^{\text{train}}$	GCR^{test}
	α	n	k_1 / k_2 / λ
SYSU-MM01	0.5	100	28 / 7 / 0.5
RegDB	0.5	50	21 / 10 / 0.4
LLCM	0.5	100	19 / 5 / 0.7

6) Synergy: The efficacy of the CDL loss hinges on the model's ability to cluster samples of the same class across different cameras. In more challenging datasets, this becomes difficult to achieve, which explains why directly replacing the InfoNCE loss with the CDL loss in the LLCM does not yield the anticipated performance improvement, as evidenced by the comparison between Strategy 2 and the Baseline. In such scenarios, the CDL loss functions more like a local smoothing loss. However, when combined with DCMU, which enhances the model's clustering capability, the CDL loss is better aligned with its intended purpose. This is illustrated by the comparison between Strategy 3 and Strategy 1, where integrating the CDL loss with DCMU results in a substantially greater performance boost than introducing the CDL loss alone. This effect is even more pronounced in the LLCM.

7) Limitation: The quality of initial clustering significantly influences the performance gains of GCR. Ablation experiments show that while GCR post-processing consistently improves performance across strategies 1* to 5*, the extent of improvement correlates positively with the initial pre-processing performance—higher initial performance leads to greater post-optimization gains. GCR exhibits computational complexity comparable to cosine distance, enabling its broad applicability in cosine-based scenarios. However, its GPU-accelerated implementation is constrained by GPU memory capacity, a critical consideration in practical applications.

D. Parameter Analysis

In this section, we examine how hyperparameters affect model performance. Table IX shows the optimal hyperparameter values for the SYSU-MM01, RegDB, and LLCM datasets. A detailed numerical analysis follows. The mAP for each dataset is calculated by averaging the results obtained from two distinct validation methodologies. And the first partition of the test set is utilized as a validation set for optimization.

1) Study the Hyperparameter of CDL Loss: The hyperparameter n represents the number of nearest negative proxies considered for each query. In this study, we examine the impact of varying values of n. As shown in Fig.6(a), the optimal value of n is 100 for both the LLCM and SYSU-MM01, and 50 for the RegDB. The performance variation primarily arises from differences in dataset scale. A larger value of n shifts the model's focus from local to global patterns, potentially compromising its effectiveness. On the other hand, an excessively small n may cause the model to overly concentrate on distinguishing challenging negative samples, neglecting other important aspects. Therefore, the selection of n should be proportional to the size of the dataset.

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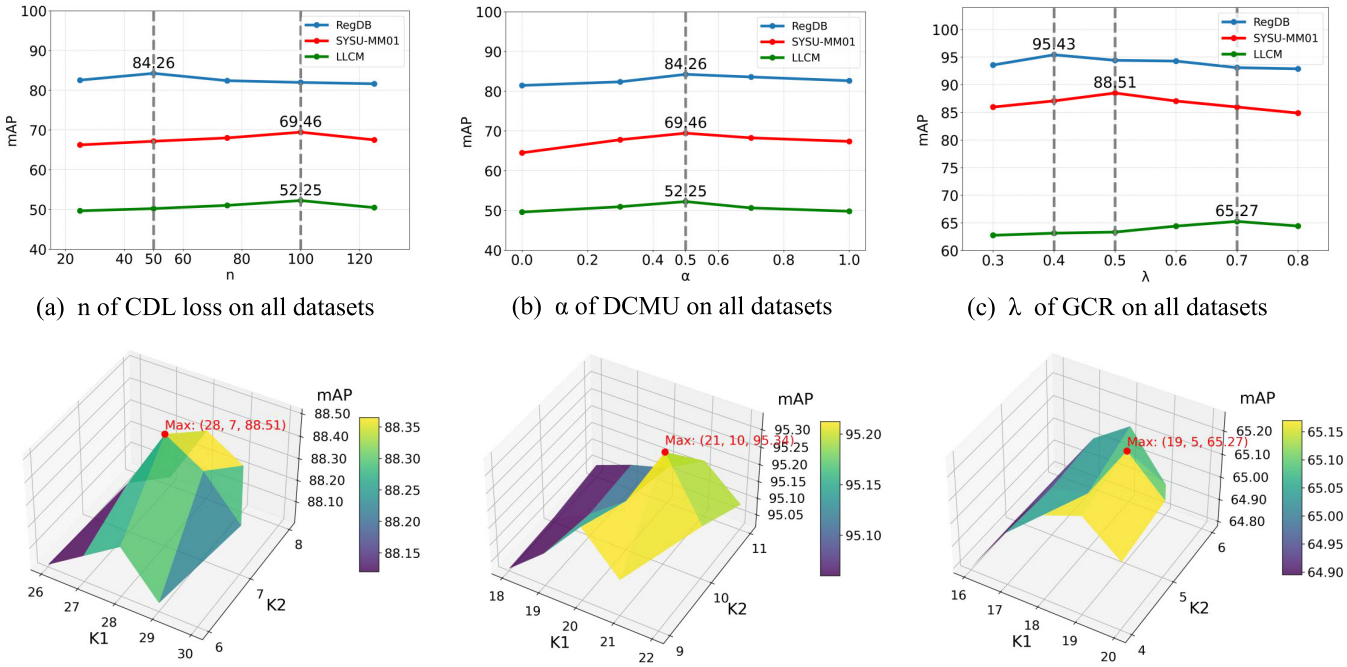

(a) n of CDL loss on all datasets (b) α of DCMU on all datasets (c) λ of GCR on all datasets

(d) k_1, k_2 of GCR on SYSU-MM01 (e) k_1, k_2 of GCR on RegDB (f) k_1, k_2 of GCR on LLCM

Fig. 6. Illustration of hyperparameters adjustment. Including the hyperparameter n for CDL loss, the hyperparameter α for DCMU, and the hyperparameters k_1, k_2, and λ for GCR, mAP is calculated by averaging across two retrieval modes, using the first partition of the test set as a validation set for optimization.

2) Study the Hyperparameter of DCMU: In DCMU, the hyperparameter α is used to balance the weighting between the most challenging samples and the less challenging ones during the update process. A higher α value increases the model's focus on difficult samples, allowing for better extraction of challenging data, but it may also lead to the neglect of simpler samples. Our experiments, as shown in Fig.6(b), indicate that setting α to 0.5 yields the best performance across all datasets.

3) Study the Hyperparameters of GCR: The hyperparameters k_1, k_2, and λ are key to optimizing GCR's performance. k_1 determines the number of nearest neighbors for initializing the distance matrix, while λ controls the strength of similarity adjustments between inter-camera and intra-camera samples. Once initialized, the graph is iteratively updated using the k_2 nearest neighbors of each data point. Thus, k_1 and λ set the initial conditions, and k_2 guides the iterative refinement. Fig.6(c) shows the impact of adjusting λ, and Fig.6(d-f) shows the effects of adjusting k_1 and k_2 together. Our experiments found the optimal testing settings to be $k_1 = 28$, $k_2 = 7$, and $\lambda = 0.5$ for SYSU-MM01; $k_1 = 21$, $k_2 = 10$, and $\lambda = 0.4$ for RegDB; and $k_1 = 19$, $k_2 = 5$, and $\lambda = 0.7$ for LLCM.

The choice of k_1 is influenced by the distribution of the probe and gallery sets in the dataset. For SYSU-MM01, with its small gallery set, a larger k_1 is necessary for stable performance. LLCM and RegDB have similar test set distributions, so their k_1 values are also similar. The selection of k_2 and λ depends on the dataset's difficulty. In more challenging datasets, where the data distribution is complex or contains significant noise, forming reliable clusters becomes more difficult. This can diminish the effectiveness of neighborhood information and may even introduce misleading information, thereby impacting the model's performance. Therefore, for datasets that do not perform as well, it is recommended to use a smaller k_2 and a larger λ to avoid interference.

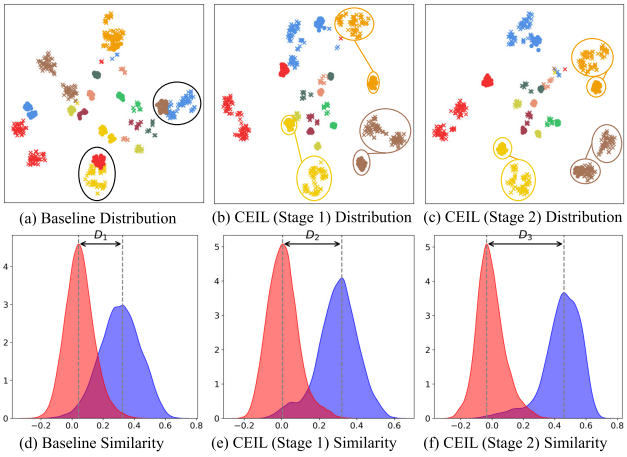

Fig. 7. Visualization of feature distribution. The first row displays t-SNE results, using circles for the visible modality and crosses for the infrared modality. The second row illustrates cosine similarity distributions, with red representing negative pairs and blue representing positive pairs.

E. Visualization

1) Feature Distribution: We selected 10 identities from the SYSU-MM01 dataset to generate t-SNE maps, shown in Fig.7(a-c) for compare the baseline method, the first stage of the CEIL method, and the second stage of the CEIL method.

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Fig. 8. Visualization of retrieval results. The leftmost image is an infrared image, while the ten images on the right are its nearest neighbors in the RGB modality. Images with a green border indicate correct recognition, whereas those with a red border indicate incorrect recognition.

The baseline method reveals noticeable clustering errors, highlighted in black boxes. In the first stage of CEIL, although the inter-class distances between different categories significantly increased, the aggregation of cross-modality samples remained suboptimal. After introducing the PCM module for cross-modality training in the second stage, the clustering performance improved markedly, particularly in the highlighted regions. However, some challenging complex categories still exhibit clustering confusion, indicating a direction for future.

We also analyzed the cosine similarity of positive and negative pairs for these models, as shown in Fig. 7(d-f). The results show that from the baseline to the first stage of CEIL, the overlap between positive and negative distributions decreased, and the distance between their centers increased significantly, indicating better discrimination. In the second stage of CEIL, the distributions separated further, with reduced overlap and an even larger center distance, demonstrating enhanced discrimination. This aligns with the t-SNE visualizations, confirming the method's effectiveness.

2) Retrieval Result: To further evaluate the effectiveness of our proposed CEIL and GCR, we obtained the retrieval results of CEIL on the SYSU-MM01 dataset and optimized them using GCR, as shown in Fig.8. Despite the significant modality differences that pose challenges to the UVI-ReID task, CEIL maintains a relatively high accuracy. By applying GCR for post-processing, the features extracted by CEIL are enhanced by incorporating prior camera information and iteratively processing on a graph network. This greatly enriches the effective information and improves cross-modal feature matching, almostly achieving the goal of accurate recognition.

V. Conclusion

This paper presents an innovative two-stage CEIL. In the first stage, we use CDL loss and the DCMU strategy to train only at the camera proxy level. This ensures the establishment of reasonable intra-modal clustering associations. In the second stage, building on the first stage, we incorporate PCM for cross-modal training at both the camera proxy and class levels, thereby creating reliable cross-modal mappings. After training, we apply GCR for post-processing to optimize the test results, with GCR recalibrating cross-modal identity matching. Extensive experiments validate the exceptional performance of our CEIL, significantly narrowing the gap between supervised and unsupervised visible-infrared person re-identification. Since CEIL operates without modifying the backbone, we intend to explore a more efficient backbone in future research.

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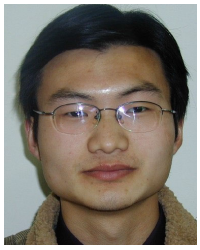
Run-Sen Xia received the bachelor's degree from Anhui University of Science and Technology in 2023. He is currently pursuing the master's degree with Anhui University. His research interests include deep learning algorithms and computer vision tasks, such as cross-modal person re-identification and multi-view truck re-identification.

Xue-Yan Wang received the master's degree from Shanghai Jiao Tong University in 2010. She is currently pursuing the Ph.D. degree in computer science and technology with Anhui University. Her research interests include artificial intelligence applications, intelligent transportation, and vehicle re-identification.



Si-Bao Chen (Member, IEEE) received the Ph.D. degree in 2006. He was a Post-Doctoral Researcher at iFLYTEK Company Ltd. He was also a Visiting Scholar at The University of Texas at Arlington from 2014 to 2015. He is currently a Professor with Anhui University, China, with a focus on pattern recognition and computer vision. He has published over 60 academic articles, with more than 50 indexed by EI/SCI, and over ten featured in CCF A class/JCR Q1 journals. He holds five granted national invention patents. He is experienced in intelligent visual analysis, particularly in transportation and public safety. He received the Best Paper Award from the BICS Conference in 2018.

Jin Tang received the B.Eng. degree in automation and the Ph.D. degree in computer science from Anhui University, Hefei, China, in 1999 and 2007, respectively. He is currently a Professor with the School of Computer Science and Technology, Anhui University. He has published over 100 academic articles, with more than 40 papers in CCF A-class conferences and JCR Q2 or higher journals. His current research interests include computer vision, pattern recognition, machine learning, and deep learning.

Bin Luo (Senior Member, IEEE) received the B.Eng. degree in electronics and the M.Eng. degree in computer science from Anhui University, Hefei, China, in 1984 and 1991, respectively, and the Ph.D. degree in computer science from the University of York, York, U.K., in 2002. From 2000 to 2004, he was a Research Associate with the University of York. He is currently a Professor with Anhui University. His research interests include graph spectral analysis, large image database retrieval, image and graph matching, statistical pattern recognition, digital watermarking, and information security.

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